

COS 314 Assignment 2

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1 Genetic Algorithm

A Genetic Algorithm is a population based metaheuristic inspired by natural selection and evolution. It maintains a population of candidate solutions that evolve over generations using selection, crossover, and mutation. Fitter individuals are more likely to pass their traits, guiding the search toward optimal solutions.

1.1 Special Relevance

The 0/1 Knapsack Problem maps naturally to binary strings. GA performs strong global exploration, avoiding premature convergence and exploring multiple combinations simultaneously.

1.2 Functionality

Representation and Initialization: Solutions are binary strings where 1 means include and 0 means exclude. Initial population is randomly generated.

Fitness and Constraint Handling: Fitness is total value. If overweight, a stochastic repair removes items randomly until feasible.

Selection: Tournament selection with moderate pressure.

Reproduction: Single-point crossover with bit-flip mutation.

Elitism: Top individuals are preserved.

1.3 Configuration

- Population Size: 100
- Generations: 500
- Crossover Rate: 0.8
- Mutation Rate: 0.02
- Tournament Size: 5
- Elitism Count: 2

2 Iterated Local Search

ILS is a trajectory-based metaheuristic that alternates between local search and perturbation to escape local optima.

2.1 Special Relevance

Knapsack landscapes are highly correlated. ILS exploits this with efficient local improvements and controlled exploration.

2.2 Functionality

Initialization: Random feasible solution.

Local Search: One-bit flip improvements until local optimum.

Perturbation: Three random bit flips.

Acceptance: Strict improvement only.

2.3 Configuration

- Iterations: 500
- Neighborhood: Single bit flip
- Perturbation Strength: 3

3 Results

Instance	Alg	Seed	Best	Optimum	Runtime
f1	ILS	22	295	293	0.005
	GA	22	293	293	0.047
f2	ILS	22	1024	1024	0.071
	GA	22	1024	1024	0.145
f3	ILS	22	35	35	0.001
	GA	22	35	35	0.035
f4	ILS	22	23	23	0.004
	GA	22	23	23	0.068
f5	ILS	22	481.1	481.1	0.012
	GA	22	481.1	481.1	0.052
f6	ILS	22	52	52	0.007
	GA	22	52	52	0.045
f7	ILS	22	107	107	0.006
	GA	22	107	107	0.060
f8	ILS	22	9767	9767	0.027
	GA	22	9767	9767	0.091
f9	ILS	22	130	130	0.005
	GA	22	130	130	0.059
f10	ILS	22	1025	1025	0.018
	GA	22	1025	1025	0.068
knapPI	ILS	22	9147	9147	0.203
	GA	22	7907	9147	0.341

4 Statistical Analysis

Wilcoxon signed-rank test ($\alpha = 0.05$).

Null hypothesis: equal performance.

Most instances (9/11) resulted in ties. Only 2 non-zero differences:

- f1: +2
- knapPI: +1240

$W^+ = 3$, $W^- = 0$, $W = 0$.

Conclusion: insufficient sample size to reject null hypothesis.

5 Analysis

ILS consistently reaches high-quality solutions faster. The knapsack landscape favors greedy improvements, which ILS exploits efficiently. GA explores broadly but struggles with fine tuning and wastes time evaluating weak candidates early.

ILS proves more efficient and reliable for these instances.